Final Exam Problems

MATH 598 / 784 · High-Dimensional Probability

November 23–27, 2026

Prepare all eight problems. The oral exam consists of two randomly selected problems presented at the blackboard in **Burnside 925**. There are **20 minutes for the entire examination**, within your booked 30-minute appointment. Arrive at the start of your appointment. The instructor may ask you to explain steps, justify assumptions, or discuss the cases specified in the problems. You may cite the facts explicitly supplied below without proving them.

Each of the two presentations is worth up to 25 points. A replacement problem costs 5 points; the total score is at least zero. For the complete procedure, rubric, and booking information, see the final exam page.

Throughout, $C, c > 0$ denote absolute constants, whose values may change. All logarithms are natural. Write $B_p^n = \{x \in \mathbb{R}^n : \|x\|_p \le 1\}$, and, for a nonempty bounded set T in a Euclidean space, write $w(T) = \mathbb{E}\sup_{t\in T}\langle g, t\rangle$ for standard Gaussian g. A standard Gaussian matrix has independent $N(0,1)$ entries; $\|A\|_F^2 = \sum_{i,j} A_{ij}^2$ denotes its squared Frobenius norm.

1. Estimating a trace without reading the matrix

Let A be a nonzero positive semidefinite $n \times n$ matrix. Suppose that we can compute Av for a chosen vector v, but cannot read the entries of A directly. Let $g \sim N(0, I_n)$ and let $\varepsilon \in \{-1, 1\}^n$ have independent uniform sign coordinates.

1. Show that $g^{\mathsf{T}} Ag$ and $\varepsilon^{\mathsf{T}} A\varepsilon$ are unbiased estimators of $\operatorname{tr} A$, and prove
$$\operatorname{Var}(g^{\mathsf{T}} Ag) = 2\|A\|_F^2, \qquad \operatorname{Var}(\varepsilon^{\mathsf{T}} A\varepsilon) = 2\sum_{i\neq j} A_{ij}^2.$$
2. Choose either estimator. Form b independent batches, each containing k independent copies, and let $\widehat{\tau}$ be a median of the b batch averages. For $0 < \delta, \rho < 1$, prove that
$$k \ge C\delta^{-2}, \qquad b \ge C\log(2/\rho) \quad \Longrightarrow \quad \mathbb{P}\{|\widehat{\tau} - \operatorname{tr} A| > \delta \operatorname{tr} A\} \le \rho.$$
Thus only $O(\delta^{-2}\log(2/\rho))$ matrix-vector products are needed, independently of n.

Hint. Use positivity of the eigenvalues to bound the variance relative to $(\operatorname{tr} A)^2$, then control how many batch averages are inaccurate.

2. Gaussian concentration after a nonlinear map

Let $A \in \mathbb{R}^{m\times n}$ be a fixed matrix whose rows have Euclidean norm one. For independent $g, h \sim N(0, I_n)$, set
$$X(g) = \tanh(Ag), \qquad \kappa = \mathbb{E}\tanh^2(Z), \quad Z \sim N(0,1), \qquad r = \frac{m}{\|A\|_{\mathrm{op}}^2},$$
where tanh acts coordinatewise.

1. Prove that $\mathbb{E}\|X(g)\|_2^2 = m\kappa$ and, for every $t > 0$,
$$\mathbb{P}\left\{\left|\frac{\|X(g)\|_2^2}{m} - \kappa\right| > t\right\} \le 2e^{-crt^2}.$$
2. Prove that, for every $t > 0$,
$$\mathbb{P}\left\{\left|\frac{\langle X(g), X(h)\rangle}{m}\right| > t\right\} \le 2e^{-crt^2}.$$
Deduce that, for $0 < \varepsilon < 1$, with probability at least $1 - 6e^{-cr\varepsilon^2}$,
$$\frac{|\langle X(g), X(h)\rangle|}{\|X(g)\|_2\|X(h)\|_2} \le \varepsilon.$$
The denominator is nonzero almost surely.

Hint. Use $|\tanh u| \le 1$, $0 \le \tanh'(u) \le 1$, and the oddness of tanh. The coordinates of $X(g)$ need not be independent.

3. The ground-state energy of a Gaussian spin system

For $n \geq 2$, let $(Z_{ij})_{1 \leq j < i \leq n}$ be independent standard normal random variables. For $\sigma \in \{-1,1\}^n$, define

$$H_\sigma = \sum_{i>j} Z_{ij}\sigma_i\sigma_j, \qquad E_n = \min_{\sigma \in \{-1,1\}^n} H_\sigma.$$

1. Prove that $\mathbb{E}E_n = -\mathbb{E}\max_\sigma H_\sigma$ and that $-\mathbb{E}E_n \leq Cn^{3/2}$.
2. Construct signs successively by taking $\sigma_1 = 1$ and choosing σ_i to maximize $\sigma_i \sum_{j<i} Z_{ij}\sigma_j$. Condition on the previous rows to show that the expected value attained is $\sqrt{2/\pi}\sum_{i=2}^n \sqrt{i-1}$. Deduce $cn^{3/2} \leq -\mathbb{E}E_n$.
3. Prove that
$$\mathbb{P}\left\{\left|\frac{E_n}{\mathbb{E}E_n} - 1\right| > \delta\right\} \leq 2e^{-c\delta^2 n} \qquad (\delta > 0).$$
In particular, $E_n/\mathbb{E}E_n \to 1$ in probability.

Facts you may use. For centered Gaussian variables with variances at most v^2, $\mathbb{E}\max_{1\leq j\leq N} X_j \leq v\sqrt{2\log N}$, without any independence assumption.

Gaussian deviation theorem. The next two problems may use the following result without proof. Let A be an $m \times n$ standard Gaussian matrix. For every compact symmetric set $S \subseteq \mathbb{R}^n$ containing 0 and every $u \geq 0$, with probability at least $1 - 2e^{-u^2}$,

$$\sup_{h\in S}\left|\|Ah\|_2 - \sqrt{m}\|h\|_2\right| \leq C_0\big[w(S) + u\,\mathrm{rad}(S)\big], \qquad \mathrm{rad}(S) = \sup_{h\in S}\|h\|_2,$$

where $C_0 > 0$ is an absolute constant.

4. Random sections of the octahedron

Let $1 \leq d < n$, let A be a $d \times n$ standard Gaussian matrix, and let $E = \ker A$. All radii and diameters below are Euclidean.

1. Put $T_r = B_1^n \cap rB_2^n$. If $n^{-1/2} \leq r \leq 1$, prove
$$w(T_r) \leq C\sqrt{\log(enr^2)}.$$
You may use $\mathbb{E}(\sum_{i=1}^s (g_i^*)^2)^{1/2} \leq C\sqrt{s\log(en/s)}$, where $g_1^* \geq \cdots \geq g_n^*$ are the decreasing rearrangement of $|g_1|, \ldots, |g_n|$.
2. Prove the Garnaev–Gluskin bound
$$\mathbb{E}\,\mathrm{diam}(B_1^n \cap E) \leq C\sqrt{\frac{\log(en/d)}{d}}.$$

Hint. Apply the Gaussian deviation theorem to T_r for a suitable r.

5. Low-rank recovery by nuclear-norm minimization

For $M \in \mathbb{R}^{p\times q}$, define its nuclear norm by $\|M\|_* = \sum_j \sigma_j(M)$, the sum of its singular values, and use $\langle B, M\rangle_F = \operatorname{tr}(B^{\mathsf{T}}M)$. Fix $1 \le r \le \min\{p,q\}$. Let $G_1, \dots, G_m$ be independent standard Gaussian $p \times q$ matrices and set

$$\mathcal{A}(M) = \big(\langle G_i, M\rangle_F\big)_{i=1}^m.$$

1. Prove that if $\operatorname{rank}(X) \le r$ and $\|X + H\|_* \le \|X\|_*$, then
$$\|H\|_* \le 2\sqrt{2r}\,\|H\|_F.$$

2. Observe $y = \mathcal{A}(X) + \eta$, where $\operatorname{rank}(X) \le r$ and $\|\eta\|_2 \le \delta$, and choose any minimizer
$$\widehat{X} \in \arg\min_M \{\|M\|_* : \ \|\mathcal{A}(M) - y\|_2 \le \delta\}.$$
Prove that if $m \ge Cr(p+q)$, then with probability at least $1 - 2e^{-cm}$,
$$\|\widehat{X} - X\|_F \le \frac{C\delta}{\sqrt{m}},$$
simultaneously for every such X and η. In particular, recovery is exact when $\delta = 0$.

Nuclear-norm facts. You may use that $\|\cdot\|_*$ is a norm, is invariant under orthogonal changes of basis, and satisfies

$$\|M\|_* \le \sqrt{\operatorname{rank}(M)}\,\|M\|_F, \qquad \sup_{\|M\|_* \le 1} \langle B, M\rangle_F = \|B\|_{\mathrm{op}}.$$

The first inequality is Cauchy–Schwarz for the singular values. For the second, write $M = \sum_j \sigma_j u_j v_j^{\mathsf{T}}$ and use $|u_j^{\mathsf{T}} B v_j| \le \|B\|_{\mathrm{op}}$; equality is attained by $M = uv^{\mathsf{T}}$ for leading left and right singular vectors of B. The nuclear norm of a block-diagonal matrix is the sum of the nuclear norms of its blocks. You may also use $\mathbb{E}\|G\|_{\mathrm{op}} \le \sqrt{p} + \sqrt{q}$ for a standard Gaussian $p \times q$ matrix G.

Hint for part 1. In singular-vector bases for X, its only nonzero block is the top-left block of size $s \times s$, where $s = \operatorname{rank}(X)$. Split

$$H = H_{\|} + H_{\perp}, \qquad H_{\|} = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & 0 \end{pmatrix}, \qquad H_{\perp} = \begin{pmatrix} 0 & 0 \\ 0 & H_{22} \end{pmatrix}.$$

Use $\operatorname{rank}(H_{\|}) \le 2r$ and compare $\|X + H_{\perp}\|_*$ with $\|X + H\|_* + \|H_{\|}\|_*$.

6. A large Euclidean section of the ℓ^1-ball

Let G be an $n \times k$ standard Gaussian matrix, where $1 \le k \le n$, and put $\mu = \mathbb{E}|g| = \sqrt{2/\pi}$ for $g \sim N(0,1)$.

1. Prove that there are absolute constants $c_0, c_1 > 0$ such that, if $k \le c_0 n$, then with probability at least $1 - 2e^{-c_1 n}$,
$$\frac{\mu}{2} n\|v\|_2 \le \|Gv\|_1 \le 2\mu n\|v\|_2 \qquad \text{for every } v \in \mathbb{R}^k.$$

2. Deduce that, for $k \le c_0 n$ (reducing the absolute constants if needed), with probability at least $1 - 2e^{-cn}$ the k-dimensional subspace $F = \operatorname{range}(G)$ satisfies
$$c\sqrt{n}\|z\|_2 \le \|z\|_1 \le \sqrt{n}\|z\|_2 \qquad \text{for every } z \in F.$$

Hint. For a fixed unit vector v, use Gaussian concentration for $\|Gv\|_1$, then pass from a sufficiently fine net to the whole sphere. You may use the Gaussian singular-value estimates from the course.

7. Generic chaining on a binary tree

For a finite metric space (T, d), define

$$\gamma_2(T,d) = \inf_{(T_j)} \sup_{t\in T} \sum_{j\ge 0} 2^{j/2} d(t, T_j), \qquad |T_0| = 1, \quad |T_j| \le 2^{2^j} \ (j \ge 1).$$

You may use **Talagrand's theorem**: for a centered Gaussian process with canonical metric $d(s,t) = (\mathbb{E}|X_s - X_t|^2)^{1/2}$,

$$c\gamma_2(T,d) \le \mathbb{E}\max_{t\in T} X_t \le C\gamma_2(T,d).$$

Let $T = \{0,1\}^m$, $m \ge 1$, be the leaves of a binary tree of depth m. Attach independent standard Gaussian variables g_e to its edges and put $X_t = \sum_{e\in[\varnothing,t]} g_e$.

1. If $\ell(s,t)$ is the length of the common prefix of two leaves, prove $d(s,t)^2 = 2(m - \ell(s,t))$.
2. Construct an admissible sequence by representing every prefix of length $\min\{m, 2^j\}$ at level $j \ge 1$ by one leaf, and taking a single leaf at level 0. Prove directly that $\gamma_2(T,d) \le Cm$.
3. Starting at the root, repeatedly follow the edge with the larger Gaussian label. Use independence of unexplored edges to show that the expected value at the resulting leaf is $m/\sqrt{\pi}$. Deduce $\gamma_2(T,d) \ge cm$ and $cm \le \mathbb{E}\max_t X_t \le Cm$.

Hint. If a, b are independent standard normals, $\max(a,b) = (a + b + |a-b|)/2$. In part 2, bound the nonzero sum by $C\sqrt{m}\sum_{2^j<m} 2^{j/2}$, keeping the level-zero term.

8. A quadratic kernel and generic chaining

Let $a_1, \dots, a_n > 0$, put $D = \operatorname{diag}(a_1, \dots, a_n)$, and define the ellipsoid $E = DB_2^n$ and its quadratic lift

$$\Phi(x) = xx^{\mathsf{T}}, \qquad \mathcal{K} = \{\Phi(x) : x \in E\}.$$

Equip $\mathcal{K}$ with the Frobenius metric, induced by $\langle A, B\rangle_F = \operatorname{tr}(A^{\mathsf{T}}B)$. Let G be an $n \times n$ standard Gaussian matrix.

1. Prove that $\langle \Phi(x), \Phi(y)\rangle_F = \langle x, y\rangle^2$. Show that $X_x = x^{\mathsf{T}}Gx$, $x \in E$, is a centered Gaussian process with covariance $\mathbb{E}X_xX_y = \langle x, y\rangle^2$, and that its canonical pseudometric satisfies

$$d(x,y)^2 = \|x\|_2^4 + \|y\|_2^4 - 2\langle x, y\rangle^2 = \|\Phi(x) - \Phi(y)\|_F^2.$$

2. Prove the sharp estimate

$$c\left(\max_i a_i\right)\left(\sum_i a_i^2\right)^{1/2} \le \gamma_2(\mathcal{K}, \|\cdot\|_F) \le C\left(\max_i a_i\right)\left(\sum_i a_i^2\right)^{1/2}.$$

You may use Talagrand's theorem from Problem 7 for compact Euclidean sets and their continuous Gaussian processes.

Hint. For the lower bound, set $H = (G + G^{\mathsf{T}})/2$ and consider a column of DHD corresponding to the largest a_i.